CSE492 ENGINEERING PROJECT
PROPOSAL FORM

Date: 29.01.2026 CSE 492 Registration Semester:

Make sure that you have registered to the CSE492 course via OBS before filling out this form.

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CSE 492 registration type (Mark one of the following choices):

☒ Regular ☐ Single Course (Tek Ders) ☐ Supplementary Course (Ek Ders)

☐ Extended

This is a group project other members are listed:
Resul ERYURT
Ata Yağız GÜNER

Project Title	YuGoDa: Food Waste Reduction Application (Backend Module)
Abstract (200 words)	This project focuses on the end-to-end software architecture and core marketplace functionality of YuGoDa, a web application designed to mitigate food waste. The primary scope includes the development of a secure and scalable full-stack environment capable of handling distinct user roles for business vendors and consumers. On the backend, the work involves constructing high-performance RESTful APIs using Java 17 (Spring Boot) and optimizing a relational database schema in PostgreSQL to manage real-time inventory tracking and order processing. The frontend development utilizes React.js and TypeScript to build a responsive single-page application (SPA), implementing Redux Toolkit for robust state management across the platform. A key technical priority is ensuring data integrity and system security, incorporating Firebase Authentication for secure social logins and session management. Additionally, this role oversees the deployment pipeline and containerization using Docker to ensure cross-platform compatibility. The final deliverable is a production-ready system that ensures the reliability and responsiveness required to support the platform's AI-driven features.

Hardware constraints:(PC, Apple, mobile phone, other device, etc.)

- PC/Mac/Linux computers with modern web browsers for accessing the web application.
- Development machines: Personal laptops for the 3 people development team.

Software constraints: (Operating System, Progr. Languages, Libraries, API's, IDE's etc)

OS: Cross platform (Windows, macOS, Linux)
Languages: JavaScript/TypeScript (React.js frontend), Java 17 (Spring Boot backend).
Libraries: React Router, Axios, Redux Toolkit, Spring, Spring Data JPA, LangChain4j.
Database: PostgreSQL.
APIs: Google Maps API, OpenAI/Claude API (AI chatbot), Firebase Auth (Social Login and Authentication).
IDE: VS Code.
DevOps: Docker, GitHub.
Cloud: AWS.

Evaluation criteria (Quality Measurements, Scalability, Error Rate, etc)

Quality: Minimum %70unit test coverage, API response time under 500ms.

Scalability: Support for 100-200 concurrent users, 3+ restaurant listings.

Error Rate: Order processing errors under 10%.

Performance: Page load time under 7 seconds, search results under 6 seconds.

AI Accuracy*:* Achieving an F1-Score of 0.75 or higher in recommendation classification; Precision and Recall metrics will be used to evaluate the accuracy of the AI-powered assistant.

<table>
<tr><td colspan="2"><u>Design Methodology Diagrams</u>
☒ Use Case Diagrams ☒ Class Diagrams ☒ Sequence Diagrams ☐ State Diagrams</td></tr>
<tr><td colspan="2"><u>Impact area</u>
☒ Health ☒ Environmental ☒ Security ☒ Social / Ethical
☒ Sustainability ☒ Productivity Other:____________________________</td></tr>
<tr><td>Project Advisor Name, Surname, Signature:</td><td>Academic Advisor Name, Surname, Signature:</td></tr>
<tr><td>Gürhan KÜÇÜK</td><td>Emin Erkan KORKMAZ</td></tr>
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